FoodForward: A Web-Based Food Donation Platform Connecting Donors, Restaurants, and Charities

Project Proposal

Course: CSE299 Junior Project
Section: 3
Group 4

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1. Abstract

Food waste and hunger continue to be major global challenges, yet there is no centralized platform that efficiently connects donors, restaurants, and charities for food donations. **FoodForward** is a web-based platform that enables users to purchase meal packages from registered restaurants and donate them directly to verified charitable organizations. The system provides donation tracking, notifications, donation history, and an innovative **"Donate Where It's Needed Most"** feature, which automatically assigns donations to charities with the highest current demand. Developed using Django, MySQL, HTML, CSS, JavaScript, Bootstrap, and GitHub, FoodForward aims to create a transparent, efficient, and accessible food donation system while reducing food waste and supporting communities in need

2. Problem Statement

Background

Food waste and food insecurity remain major global challenges. Although many restaurants and individuals are willing to support charitable causes, there is no centralized platform that efficiently connects donors, restaurants, and charities. Existing donation methods are often manual and lack transparency, making it difficult to track donations and ensure fair distribution among charities. This highlights the need for a streamlined, transparent, and accessible food donation system.

Objectives

The proposed system aims to:

- Develop a centralized food donation platform.
- Connect donors, restaurants, and charities through one system.
- Provide transparent donation tracking.
- Reduce food waste by encouraging restaurant participation.
- Ensure fair donation distribution using an intelligent allocation feature.
- Maintain complete documentation and version control through GitHub.

3. Proposed Solution

FoodForward will be developed as a responsive web application consisting of four different user roles.

Donor

- Register and log in.
- Browse participating restaurants.
- Select meal packages.
- Choose a charity or select **Donate Where It's Needed Most**.
- Complete a simulated payment.
- Track donation history.
- Receive donation notifications.

Restaurant

- Manage restaurant profile.
- Add and edit meal packages.
- Accept donation orders.
- Update order status (Pending, Accepted, Preparing, Delivered).

Charity

- Register with administrator approval.
- Receive donated meals.
- Confirm successful delivery.
- View donation records.

Administrator

- Verify restaurants and charities.
- Manage users.
- Monitor donations.
- Generate reports and statistics.

Core Features

- User Authentication
- Restaurant Management
- Charity Management
- Meal Package Management
- Donation Tracking
- Notification System
- Simulated Payment Module
- Donation History
- Administrative Dashboard

Innovative Feature

Unlike traditional donation systems, FoodForward introduces **Donate Where It's Needed Most**, allowing the system to automatically allocate donations to charities with the highest number of pending requests. This promotes fairness and ensures resources are distributed efficiently.

4. Methodology

The project will follow the Software Development Life Cycle (SDLC) using an iterative development approach.

Technology Stack

Component	Technology
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	Django (Python)
Database	MySQL
Database Tool	phpMyAdmin
Version Control	Git & GitHub
IDE	Visual Studio Code

System Workflow

1. User registers and logs into the system.
2. User browses available restaurants.
3. User selects a meal package.
4. User chooses a charity or selects "Donate Where It's Needed Most."
5. User completes the simulated payment.
6. Restaurant receives the donation order.
7. Restaurant prepares and delivers the meal.
8. Charity confirms successful receipt.
9. Donor receives a notification confirming completion.

Database Design

The proposed database will contain the following primary tables:

- Users
- Restaurants
- Charities
- Meal Packages
- Donations
- Notifications
- Payments

The relationships between these tables will ensure efficient data storage and retrieval while maintaining data integrity.

5. Expected Outcomes

Upon completion, **FoodForward** will provide a transparent and efficient platform for food donations while connecting donors, restaurants, and charities.

Expected Outcomes:

- Reduce food waste.
- Improve donation accessibility.
- Ensure transparent donation tracking.
- Promote fair distribution of meals.
- Strengthen skills in Django, MySQL, GitHub, and full-stack web development.

.6. Future Scope

Future versions of FoodForward may include:

- Real online payment gateway integration (SSLCommerz, bKash, Nagad).
- Mobile application development.
- GPS-based delivery tracking.
- Real-time notifications using WebSockets.
- AI-based recommendation system for donation allocation.
- Email and SMS notification services.
- Analytics dashboard with data visualization.

7. Conclusion

FoodForward aims to reduce food waste and improve food donation management by providing a centralized platform connecting donors, restaurants, and charities. The system promotes transparency, accountability, and fair distribution of donations through modern web technologies. The project is feasible within the given timeline and will enhance the team's practical skills in Django, full-stack web development, database management, and collaborative software development using GitHub.

References

1. Django Software Foundation. *Django Documentation*. https://docs.djangoproject.com/
2. Oracle Corporation. *MySQL Documentation*. https://dev.mysql.com/doc/
3. Bootstrap Team. *Bootstrap Documentation*. https://getbootstrap.com/docs/
4. GitHub Docs. *GitHub Documentation*. https://docs.github.com/
5. United Nations Sustainable Development Goal 2: Zero Hunger.
6. United Nations Sustainable Development Goal 12: Responsible Consumption and Production.